

§1.565 pullback_of_surjective_is_pushout_Set

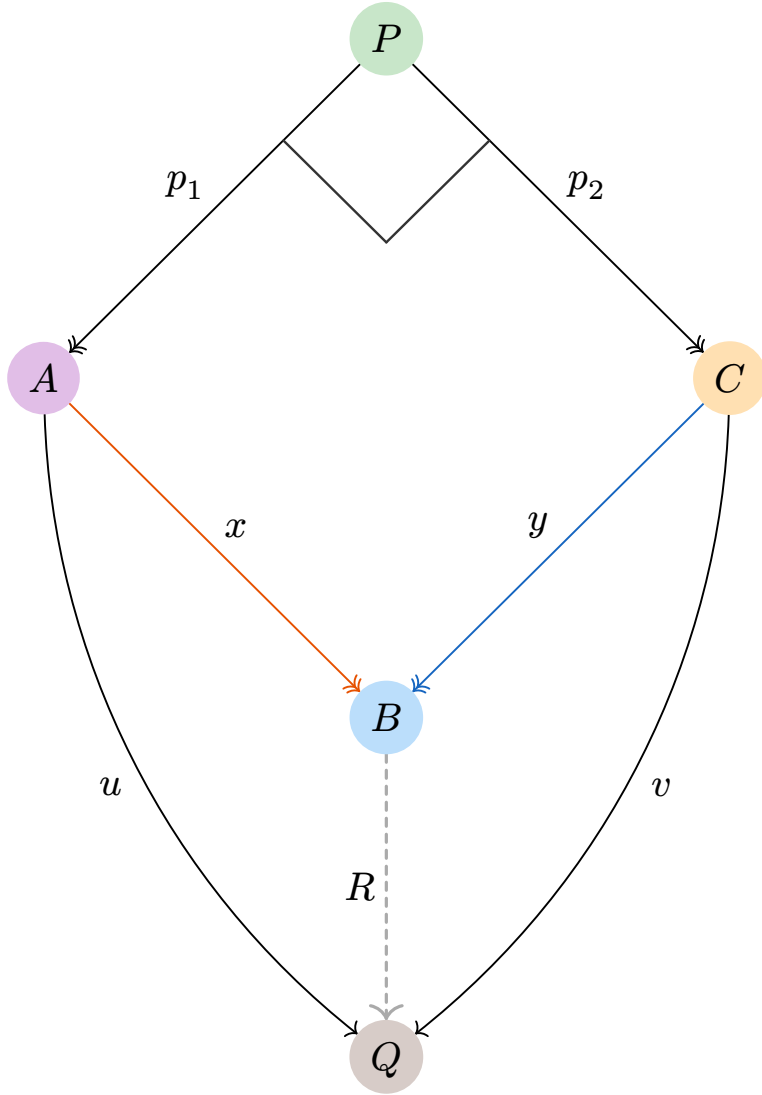


Figure 1: Pullback of covers x, y (surjections in **Set**); p_1, p_2 are covers too (pullbacks transfer covers). Given a cocone u, v with $p_1 u = p_2 v$, define the **relation** $R := x^\circ u \cap y^\circ v : B \nrightarrow Q$ – to be shown a map.

In **Set**: $b R q \iff (\exists a. x(a) = b \wedge u(a) = q) \wedge (\exists c. y(c) = b \wedge v(c) = q)$

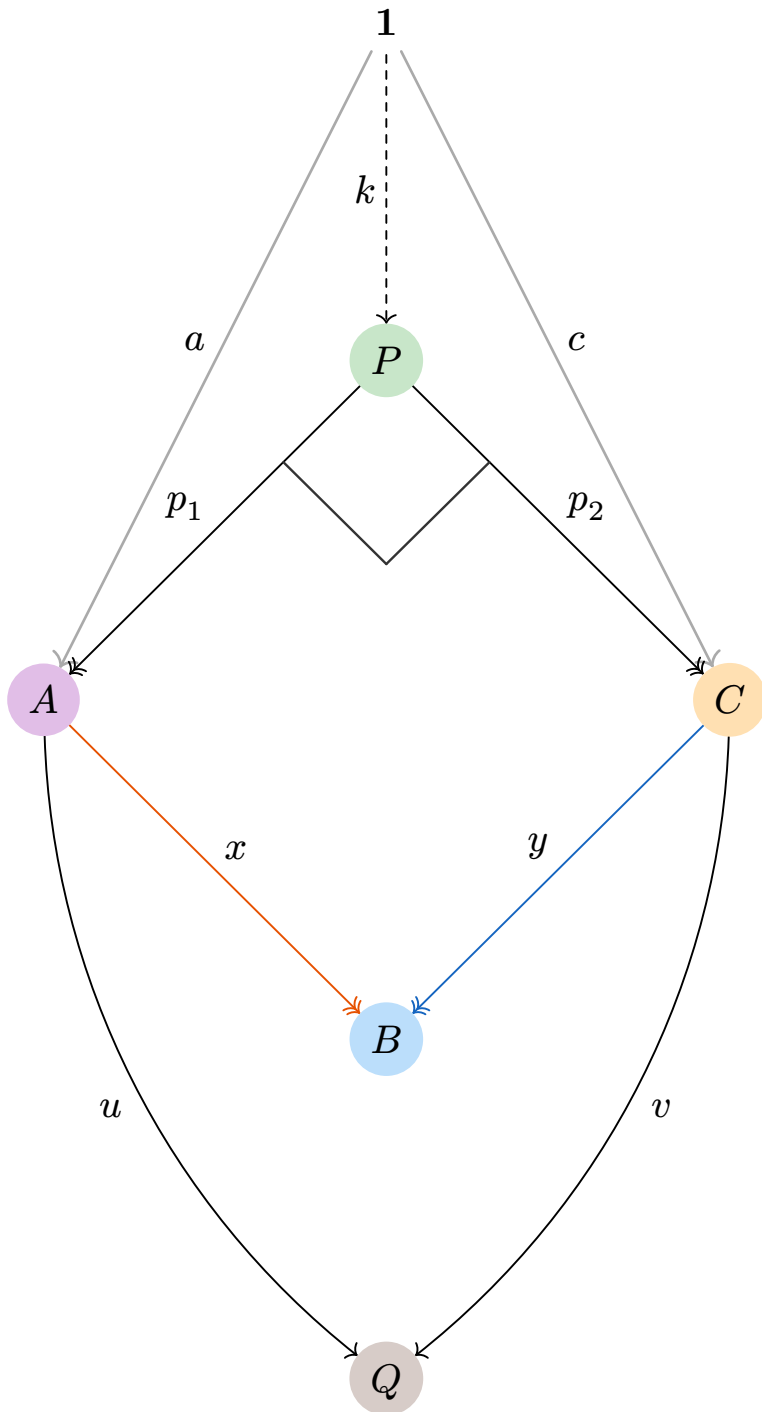
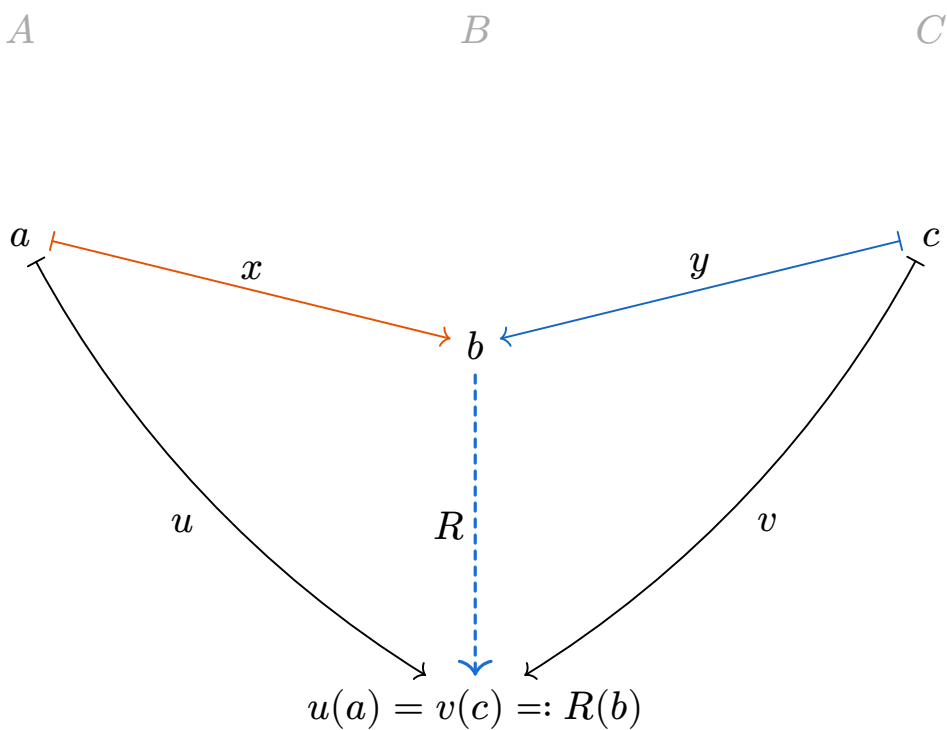


Figure 2: Key lemma: $x(a) = y(c) \Rightarrow u(a) = v(c)$. The cone $\langle a, c \rangle : \mathbf{1} \rightarrow A \times C$ lifts uniquely to $k : \mathbf{1} \rightarrow P$ with $p_1 k = a, p_2 k = c$.

$u(a) = u(p_1(k^*)) = v(p_2(k^*)) = v(c)$
(middle step: the cocone condition $p_1 u = p_2 v$)



$u(a) = v(c) =: R(b)$

Figure 3: R is a map: every b has $a \in x^{-1}(b), c \in y^{-1}(b)$ (covers), and the key lemma makes all candidate values agree, so $R(b) := u(a)$ is forced.

entire: $b R u(a)$ – witnesses a, c , with $v(c) = u(a)$ by the key lemma
simple: $b R q, b R q' \Rightarrow q = u(a) = v(c') = q'$ – the key lemma **crosses** the two halves of R

$x R = u$: $(x(a)) R (u(a))$ holds, so $R(x(a)) = u(a)$ by simplicity; likewise $y R = v$
uniqueness: x cover $\Rightarrow$ epi: $R'(x(a)) = u(a)$ already determines any other map R'

Legend:
->> hollow head = cover (surjection in **Set**),
|-> = element assignment,
dashed gray = relation,
dashed blue = the resulting map,
⌞ = pullback corner.